ORF522 HW3

(Q.1)

$$\min f(x)$$
$$\text{s.t. } g_j(x) \le 0, \quad j = 1, \ldots, m$$

- f, g_j are continuously differentiable
- x^* is local optimal solution

$\Rightarrow \nabla L(x^*) = 0$ where $L(x) = f(x) + \sum_{j=1}^{m} \lambda_j g_j(x) \mid \lambda_j \geq 0$

- $\nabla g_1(x^*), \ldots, \nabla g_m(x^*)$ are lin. indep $\Rightarrow$ satisfy LICQ

How? $A(x) = \{j, g_j(x) = 0\} \subseteq \{1, 2, \ldots, m\}$ → set

So, $\{\nabla g_j(x), j \in A(x)\} \subseteq \{\nabla g_1(x), \ldots, \nabla g_m(x)\}$
$= LI.$

So, x^* = local opt^m $\Rightarrow$ local min & LICQ holds $\Rightarrow$

① Stnt :- $\nabla f(x^*) + \sum_{j=1}^{m} \lambda_j \nabla g_j(x^*) = 0$

② PF :- $g_j(x^*) \le 0$; $j = 1$ to m } — ④.

③ DF : $\lambda_j \geq 0$; $j = 1$ to m

④ CS : $\lambda_j g_j(x^*) = 0$; $j = 1$ to m

complementary slackness

Now, let $\boxed{h_j(x) = g_j(x^*) + \nabla g_j(x^*)^T (x - x^*)}$; $j = 1$ to m

$$= \nabla g_j(x^*)^T x + c \rightarrow \text{const (indt of } x)$$

$\Rightarrow h_j(x)$ is affine & hence <u>convex</u> ($\forall j = 1$ to m)

(How? $\Rightarrow \nabla h_j(x) = \nabla g_j(x^*)$, $\nabla^2 h_j(x) = 0 \succeq 0$,

also $\text{dom}(h_j) = \mathbb{R}^n$ = convex).

Further, let $\boxed{v(x) = f(x^*) + \nabla f(x^*)^T (x - x^*)}$

$$= \nabla f(x^*)^T x + c'$$

$\Rightarrow v(x)$ = affine & hence convex

So, $\min\ f(x^*) + \nabla f(x^*)^T (x - x^*)$

s.t. $g_j(x^*) + \nabla g_j(x^*)^T (x - x^*) \le 0$, $j = 1$ to m $\Longleftrightarrow$ $\boxed{\begin{array}{l} \min\ v(x) \\ \text{s.t. } h_j(x) \le 0,\ j = 1 \text{ to } m \end{array}}$ → (P)

(v, h_j = convex $\forall j$)

$\Updownarrow$ (**)

local min = global min

Is x^* local min of ~~v(x)~~ problem (P)?

① PF: $h_j(x^*) = g_j(x^*) \le 0$ $\forall j = 1$ to m

② DF: $\lambda_j \ge 0$ $\forall j = 1$ to m (mtplied to $h_j(x)$) } → λ_j from (*)

③ CS: $\lambda_j \cdot h_j(x^*) = \lambda_j \cdot g_j(x^*) = 0$ $\forall j = 1$ to m

④ Stn^t: $\nabla v(x^*) + \sum_{j=1}^{m} \lambda_j \nabla h_j(x^*)$

$$= \nabla f(x^*) + \sum_{j=1}^{m} \lambda_j \nabla g_j(x^*)$$

$$= 0 \quad (\text{from } (*))$$

$\because$ P is convex, KKT condn satisfied $\Rightarrow x^*$ = local min

using (**), x^* = global min = global optimal

- QED

Way 2

Using subdifferential, we know that :-

if ϕ is convex, then by fermat's optimality cond^n,

$x^* =$ global minimizer $\longleftrightarrow 0 \in \partial\phi(x^*)$

~~If f is differentiable~~

~~& f is differentiable at~~ so

(we use here that f is convex)

~~So, 0 = global minimizer $\Leftrightarrow$ 0 $\in$~~

In our case, $\phi(x) = f(x) + \lambda\|x\|_1$

Note that norms are always convex

$\Rightarrow \because f(x)$ is convex, $f(x) + \lambda\|x\|_1$ is convex $(\lambda \geq 0)$

$\Rightarrow \phi(x)$ is convex.

So, $0 =$ global minimizer $\longleftrightarrow 0 \in \partial\phi(0)$

$0 =$ optimal $\longleftrightarrow 0 \in \partial\phi(0)$

$$0 \in \nabla f(0) + \lambda\, \partial\|x\|_1 \Big|_{x=0}$$

Now, from the slides,

$$\|x\|_1 = \max_{\|z\|_\infty \leq 1} z^T x$$

$$\partial\|x\|_1 = \underset{\|z\|_\infty \leq 1}{\operatorname{argmax}}\ z^T x \qquad \Rightarrow \partial\|x\|_1\Big|_{x=0} = \underbrace{[-1,1]^n}_{[-1,1]\times[-1,1]\times\cdots\times[-1,1]}$$

$$= \begin{cases} \{-1\} & ; x_i < 0 \\ [-1,1] & ; x_i = 0 \\ \{1\} & x_i > 0 \end{cases}$$

~~$\Rightarrow 0 \in \nabla f(0) + \lambda\, \partial\|x\|_1 \Rightarrow$~~

~~$\Rightarrow 0 \in \nabla f(0) + \{\{-\lambda\};$~~

So, $\lambda \, \partial \|x\|_1 \big|_{x=0} = [-\lambda, \lambda]^n$

$$\Rightarrow \quad \nabla f(0) + \lambda \, \partial \|x\|_1 \big|_{x=0} = [-\lambda + \nabla f(0)_1, \lambda + \nabla f(0)_1] \times [-\lambda + \nabla f(0)_2, \lambda + \nabla f(0)_2] \times \cdots \times [-\lambda + \nabla f(0)_n, \lambda + \nabla f(0)_n] \quad (*)$$

$0 \in (*)$ means

$$-\lambda + \nabla f(0)_1 \le 0 \le \lambda + \nabla f(0)_1,$$
$$-\lambda + \nabla f(0)_2 \le 0 \le \lambda + \nabla f(0)_2,$$
$$\vdots$$
$$-\lambda + \nabla f(0)_n \le 0 \le \lambda + \nabla f(0)_n$$

$$\Rightarrow \quad -\lambda + \nabla f(0)_i \le 0 \le \lambda + \nabla f(0)_i \quad \forall i$$

$$\Rightarrow \quad -\lambda \le -\nabla f(0)_i \le \lambda \quad \forall i$$

$$\Rightarrow \quad \lambda \ge \nabla f(0)_i \ge -\lambda \quad \forall i$$

$$\Rightarrow \quad |\nabla f(0)_i| \le \lambda \quad \forall i$$

$$\Rightarrow \max_{i=1 \text{ to } n} |\nabla f(0)_i| \le \lambda$$

$$\Rightarrow \quad \|\nabla f(0)\|_\infty \le \lambda$$

So, $0 =$ optimal $\longleftrightarrow \|\nabla f(0)\|_\infty \le \lambda$

— QED

(b) Yes, we used the fact that f is convex & we have to use this.

Suppose f is convex were not required. Then, the equivalence would hold for non convex f^n too.

But consider $f = -x^2$ $(n=1)$ & $\lambda = 0$.

So, $\|\nabla f(0)\|_\infty = |\nabla f(0)| = |-2x|_{x=0}| = |0| = 0 \leq 0$

But $x=0$ is not optimal for $\phi(x) = -x^2 + 0\|x\|_1 = -x^2$.

Infact, for $\phi(x) = -x^2$, the optimal solution does NOT exist

$\Rightarrow$ Equivalence does not hold.

So, $f^n \neq$ convex $\rightarrow$ Equivalence does not hold

So, Equivalence holds $\rightarrow$ f has to be convex.

Where did we use it? In fermat's optimality cond^n.

To show:- $\|x\|_1$ is convex.

i) dom $= \mathbb{R}^n =$ convex

ii) $\|\lambda x + (1-\lambda) y\|_1 \underset{\text{by def}^n \text{ of } \|\cdot\|_1}{\leq} \lambda \|x\|_1 + (1-\lambda)\|y\|_1 \Rightarrow \|x\|_1$ is convex.

(Q.3) $\min f_0(x)$ s.t. $f_i(x) \le 0$; where $f_i : \mathbb{R}^n \to \mathbb{R}$, $i = 0, 1, \ldots, m$ are convex $\longrightarrow$ (P1)

Let $\Phi : \mathbb{R} \to \mathbb{R}$ is an increasing & convex function. Then,

$$\boxed{P2 = \min \tilde{f}_0(x) = \Phi(f_0(x)) \quad \text{s.t. } f_i(x) \le 0,\ i = 1, \ldots, m}$$

is convex & eq^n to P1 $\downarrow$ same set of opt sol

To explore: connections b/w duals of two problem when f_i are differentiable & $\Phi(a) = \exp(a)$

$$\boxed{P1 = \min f_0(x) \quad \text{s.t. } f_i(x) \le 0} \qquad \Downarrow \qquad \boxed{P2 = \min \exp(f_0(x)) \quad \text{s.t. } f_i(x) \le 0}$$

$\exp(f_0(\lambda x + (1-\lambda) y))$
$\because f_0$ is cvx $\le \exp(\lambda f_0(x) + (1-\lambda) f_0(y))$
$\le \lambda \exp(f_0(x)) + (1-\lambda)\exp(f_0(y))$
$\because \exp(\cdot)$ is convex
$\forall \lambda \in [0,1]$ & $x, y \in \mathrm{dom}(f)$
$\Downarrow$
$\exp(f_0(\cdot)) = $ cvx
$\Downarrow$

$i = 1$ to m

$f_0, f_1, \ldots, f_m$ are convex $\Rightarrow$ P1, P2 convex

(a) Given $\lambda \in \mathbb{R}^m$ = fsb for dual of (1) $\Rightarrow \lambda \ge 0 \Rightarrow \lambda_i \ge 0\ \forall\ i = 1$ to m

& $\bar{x}$ minimizes $f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) = \underbrace{L_1(x)}_{\text{Lagrange for P1}} \Rightarrow$ (1) $\nabla L_1(\bar{x}) = 0$, (2) $\nabla^2 L_1(\bar{x}) \succeq 0$ (psd)

||ly, $L_2(x) = \exp(f_0(x)) + \sum_{i=1}^{m} \bar{\lambda}_i f_i(x)$

$\Rightarrow$ (1) $\nabla L_1(\bar{x}) = 0$

$$\Rightarrow \nabla f_0(\bar{x}) + \sum_{i=1}^{m} \lambda_i f_i(\bar{x}) = 0$$

$$\Rightarrow \exp(f_0(\bar{x})) \nabla f_0(\bar{x}) + \sum_{i=1}^{m} (\exp(f_0(\bar{x})) \lambda_i) f_i(\bar{x}) = 0$$

$$\Rightarrow \nabla \exp(f_0(\bar{x})) + \sum_{i=1}^{m} (\exp(f_0(\bar{x})) \lambda_i) f_i(\bar{x}) = 0$$

$$\Rightarrow \nabla \exp(f_0(\bar{x})) + \sum_{i=1}^{m} \tilde{\lambda}_i f_i(\bar{x}) = 0$$

$\bar{\lambda}_i = \exp(f_0(\bar{x})) \lambda_i = > 0 \cdot \ge 0 = \ge 0$

$\bar{\lambda}_0 \ge 0$

$\Rightarrow \bar{\lambda}$ = dual fsb

~~$\Rightarrow \bar{x}$ minimizes~~

$$\Rightarrow \boxed{\nabla L_2(\bar{x}) = 0}$$

② $\nabla^2 L_1(\bar{x}) \succeq 0$

$$\Rightarrow \nabla^2 f_0(\bar{x}) + \sum_{i=1}^{m} \lambda_i \nabla^2 f_i(\bar{x}) \succeq 0 \quad \text{— ①}$$

Now, $L_2(x) = \exp(f_0(x)) + \sum_{i=1}^{m} \bar{\lambda}_i f_i(x)$

$$\nabla L_2(x) = \exp(f_0(x)) \nabla f_0(x) + \sum_{i=1}^{m} \bar{\lambda}_i \nabla f_i(x)$$

$$\nabla^2 L_2(x) = \exp(f_0(x)) \nabla^2 f_0(x) + \exp(f_0(x)) \nabla f_0(x) \nabla f_0^T(x) + \sum_{i=1}^{m} \bar{\lambda}_i \nabla^2 f_i(x)$$

$$\nabla^2 L_2(\bar{x}) = \exp(f_0(\bar{x})) \nabla^2 f_0(\bar{x}) + \exp(f_0(\bar{x})) \nabla f_0(\bar{x}) \nabla f_0^T(\bar{x}) + \sum_{i=1}^{m} \bar{\lambda}_i \nabla^2 f_i(\bar{x})$$

$$= \exp(f_0(\bar{x})) \nabla^2 f_0(\bar{x}) + \sum_{i=1}^{m} \bar{\lambda}_i \nabla^2 f_i(\bar{x}) + \exp(f_0(\bar{x})) \nabla f_0(\bar{x}) \nabla f_0^T(\bar{x})$$

$$= \exp(f_0(\bar{x})) \nabla^2 f_0(\bar{x}) + \sum_{i=1}^{m} \exp(f_0(\bar{x})) \nabla^2 f_i(\bar{x}) + \exp(f_0(\bar{x})) \nabla f_0(\bar{x}) \nabla f_0^T(\bar{x})$$

$$= \exp(f_0(\bar{x})) \left(① + \nabla f_0(\bar{x}) \nabla f_0^T(\bar{x}) \right)$$

Now, ① $\succeq 0$

Let $\nabla f_0(\bar{x}) \nabla f_0^T(\bar{x}) = H$

$$v^T H v = v^T \nabla f_0(\bar{x}) \nabla f_0^T(\bar{x}) v = \|\nabla f_0^T(\bar{x}) v\|_2^2 \geq 0 \quad \forall v$$

$$\Rightarrow H \succeq 0$$

$$\Rightarrow ① + H \succeq 0$$

$$\Rightarrow \because \exp(f_0(\bar{x})) > 0, \quad \boxed{\nabla^2 L_2(\bar{x}) \succeq 0}$$

$\Rightarrow \bar{x}$ minimizes $L_2(x) = \exp(f_0(x)) + \sum_{i=1}^{m} \tilde{\lambda}_i f_i(x)$

when $\tilde{\lambda}_i = \exp(f_0(\bar{x})) \lambda_i$, $i = 1$ to m

– QED

(b) ~~If $f_0(\bar{x}) \geq 0$, $\tilde{g}(\tilde{\lambda})$ (or $\tilde{g}(\tilde{\lambda})$) is better lowerbound than $g(\lambda)$.~~

~~However~~ $p^* \geq g(\lambda) \overset{\because \exp(\cdot) \text{ is } \uparrow}{\Rightarrow} \exp(p^*) \geq \exp(g(\lambda))$ &

$\exp(p^*) \geq \tilde{g}(\tilde{\lambda})$,

we should compare $\boxed{\tilde{g}(\tilde{\lambda}) \text{ with } \exp(g(\lambda))}$

~~So, $\tilde{g}(\tilde{\lambda}) = \inf$~~ Let :-

$$\tilde{g}(\tilde{\lambda}) = g_2(\lambda) = \inf_x \left[\exp(f_0(x)) + \sum_{i=1}^m \exp(f_0(x))\, \lambda_i f_i(x)\right]$$

$$\exp(g(\lambda)) = g_1(\lambda) = \exp\left(\inf_x \left[f_0(x) + \sum_{i=1}^m \lambda_i f_i(x)\right]\right)$$

$\because$ exp is $\uparrow$

$$= \inf_x \left[\exp\left(f_0(x) + \sum_{i=1}^m \lambda_i f_i(x)\right)\right]$$

$$= \inf_x \left[\exp(f_0(x)) \cdot \exp\left(\sum_{i=1}^m \lambda_i f_i(x)\right)\right]$$

Now, since $\bar{x}$ is minimizer for the expressions,

$$g_2(\lambda) = \exp(f_0(\bar{x})) + \sum_{i=1}^m \exp(f_0(\bar{x}))\, \lambda_i f_i(\bar{x})$$

$$= \exp(f_0(\bar{x})) + \exp(f_0(\bar{x})) \sum_{i=1}^m \lambda_i f_i(\bar{x})$$

$$= \exp(f_0(\bar{x}))\left(1 + \sum_{i=1}^m \lambda_i f_i(\bar{x})\right) \quad — (2)$$

$$g_1(\lambda) = \inf \exp\left(f_0(\bar{x}) + \sum_{i=1}^m \lambda_i f_i(\bar{x})\right)$$

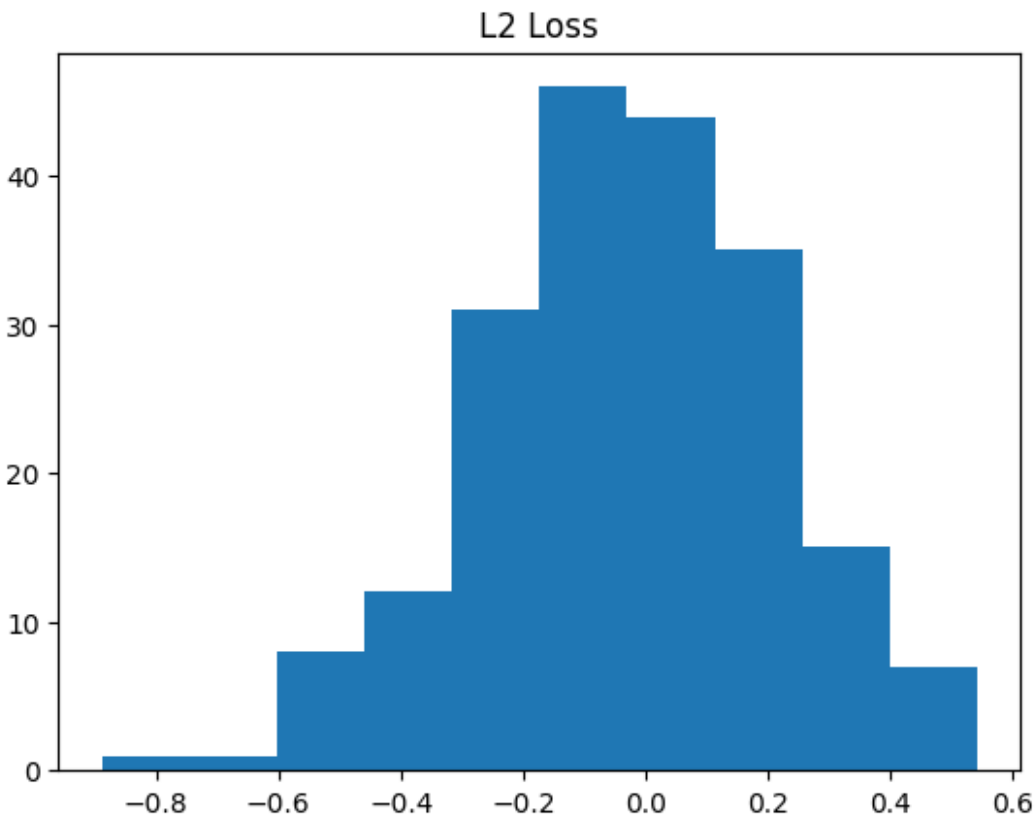
$$= \exp(f_0(\bar{x}))\, \exp\left(\sum_{i=1}^m \lambda_i f_i(\bar{x})\right) \quad — (1)$$

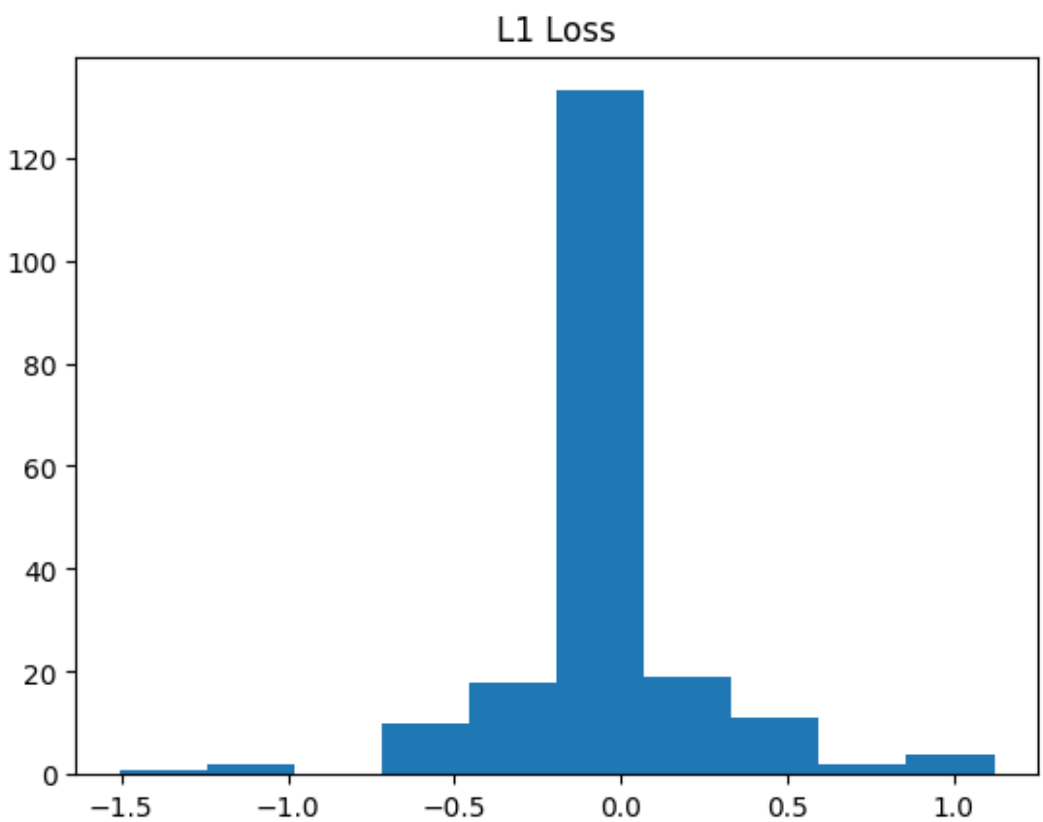
$\because \exp(☺) \geq 1 + ☺ \;\; \forall\; ☺$,

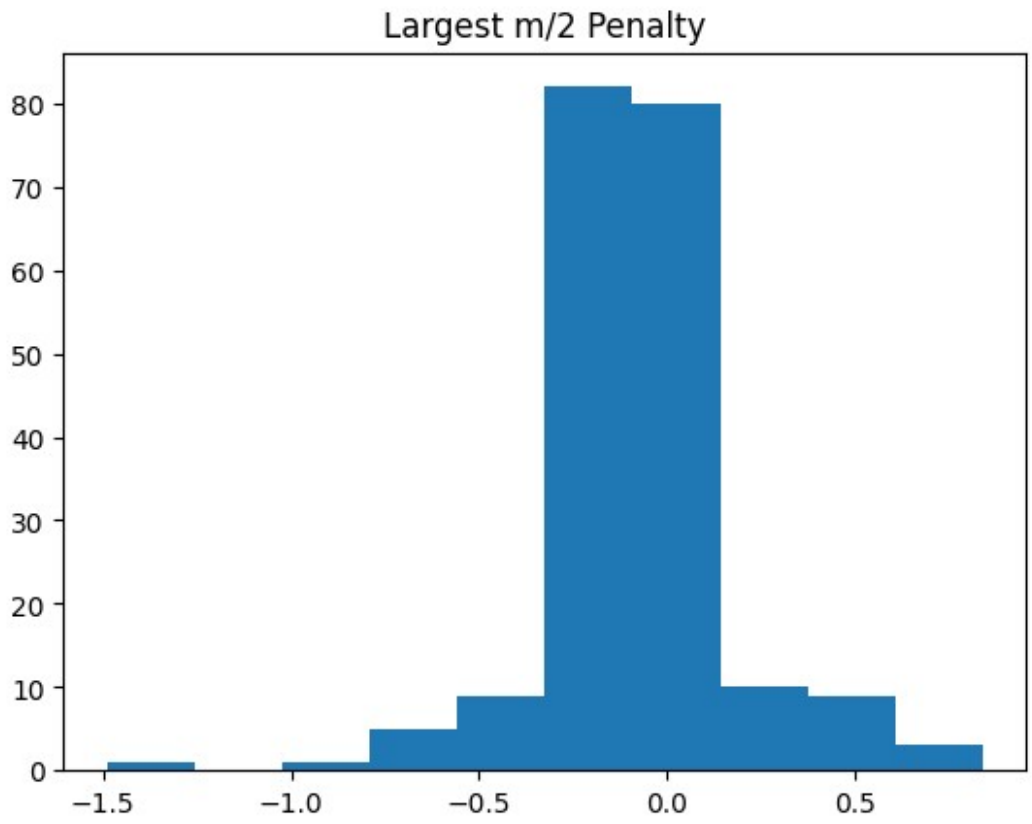
$(1) \geq (2) \Rightarrow g_1(\lambda) \geq g_2(\lambda) \Rightarrow \exp(g(\lambda)) \geq \tilde{g}(\tilde{\lambda})$

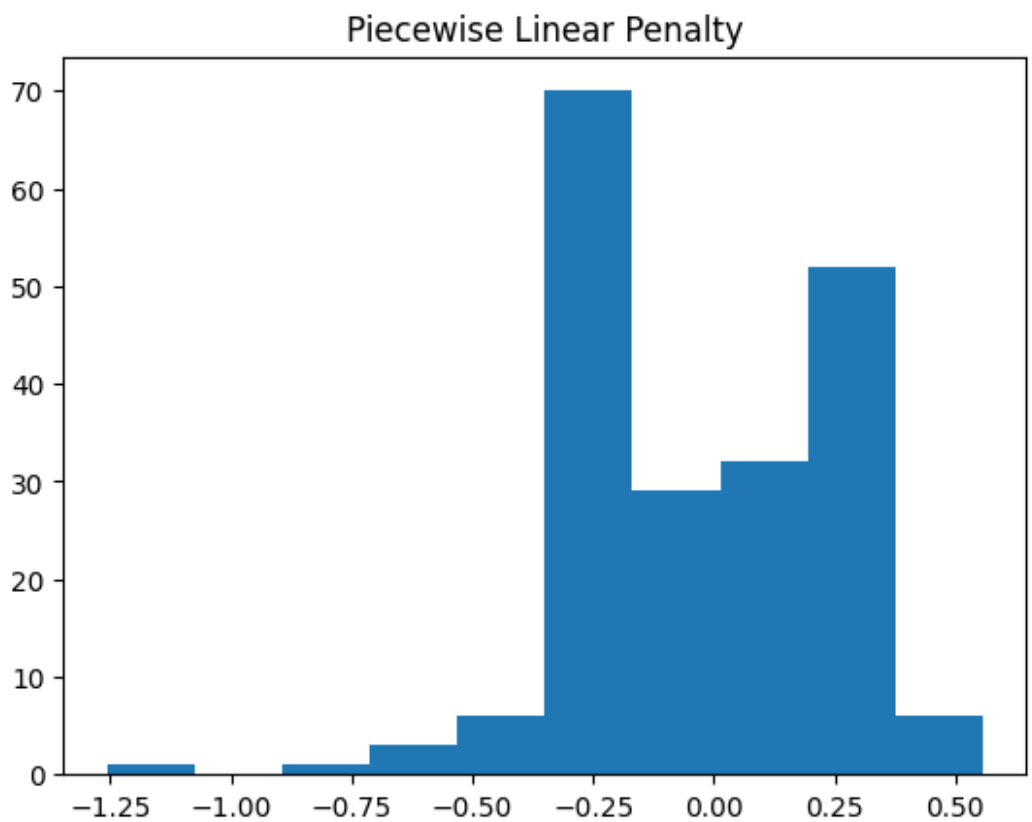
$\Rightarrow \exp(g(\lambda))$ is a better lower bound.

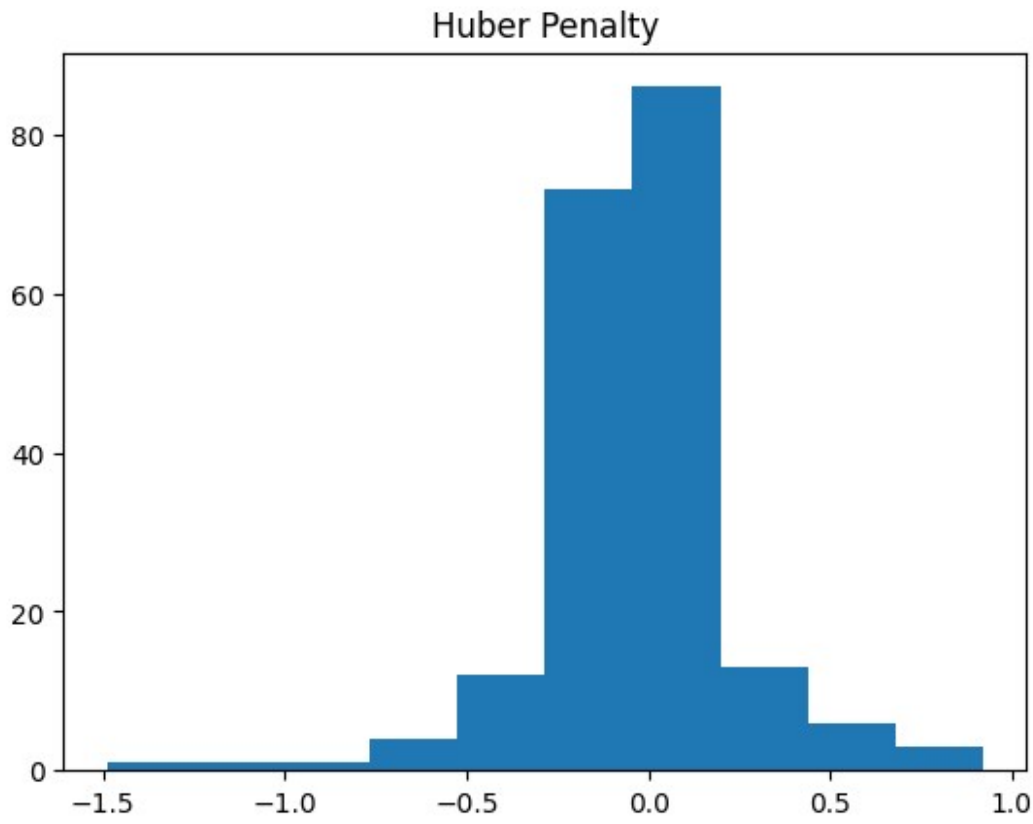
```python
import cvxpy as cp
import math
import matplotlib.pyplot as plt
import numpy as np

m = 200
n = 100
A = np.random.randn(m, n)
b = np.random.randn(m)
b = b / (1.01 * np.max(np.abs(b)))

function_vec = [lambda x: cp.norm(x, 2),
                lambda x: cp.norm(x, 1),
                lambda x: cp.sum_largest(cp.abs(x), m//2),
                lambda x: cp.sum(cp.maximum(cp.abs(x) - 0.2, 0) +
cp.maximum(cp.abs(x) - 0.3, 0)),
                lambda x: cp.sum(cp.huber(x, 0.2)),
                lambda x: cp.sum(-cp.log(1-x**2))]

num_functions = len(function_vec)
resid_vec = np.zeros((num_functions, m))
for i in range(num_functions):
    func = function_vec[i]
    # Define variables
    x = cp.Variable(n)
    # Define objective
    loss = func(A@x - b)
    objective = cp.Minimize(loss)
    # Define constraints
    constraints = []
    # Define problem
    problem = cp.Problem(objective, constraints)
    # Solve problem
    problem.solve()
    xval = x.value
    resid_vec[i, :] = A@xval - b
    # plt.hist(resid_vec[i])
    # plt.show()

func_name = ["L2 Loss", "L1 Loss", "Largest m/2 Penalty", "Piecewise
Linear Penalty", "Huber Penalty", "Log Penalty"]
for i in range(num_functions):
    plt.hist(resid_vec[i])
    plt.title(func_name[i])
    plt.show()
```

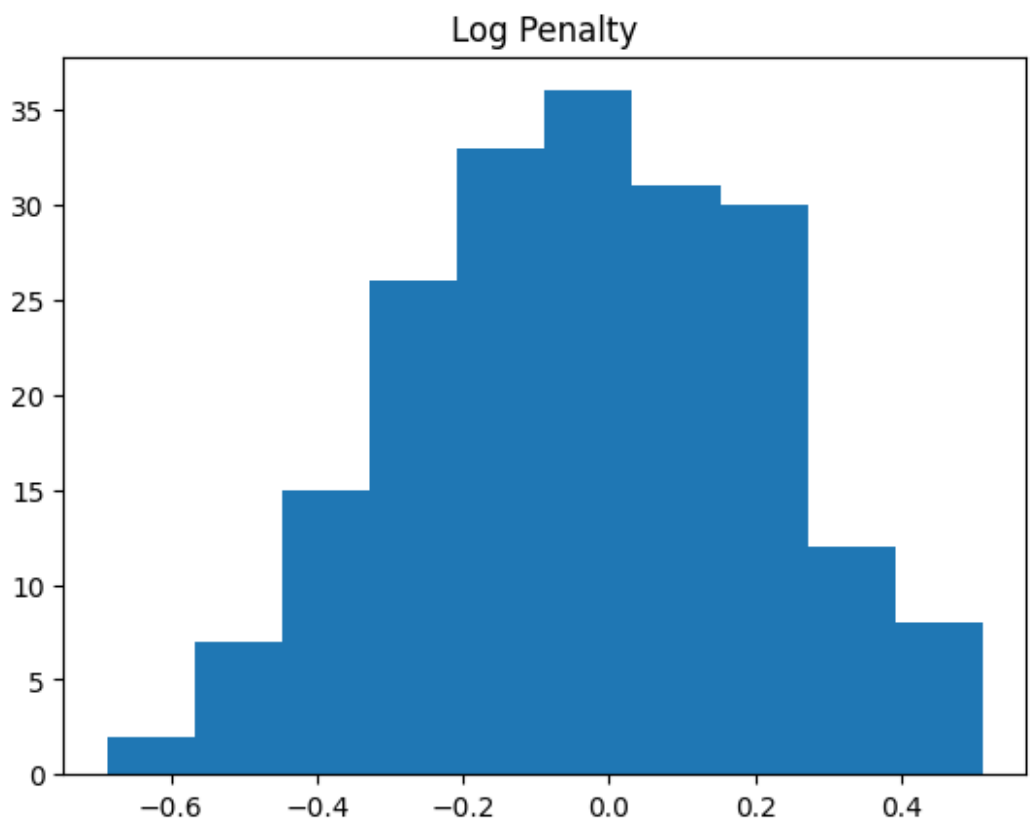

L2 Loss
0
10
20
30
40
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6


L1 Loss
0
20
40
60
80
100
120
−1.5
−1.0
−0.5
0.0
0.5
1.0


Largest m/2 Penalty
80
70
60
50
40
30
20
10
0
−1.5
−1.0
−0.5
0.0
0.5


Piecewise Linear Penalty
70
60
50
40
30
20
10
0
−1.25
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50


Huber Penalty
80
60
40
20
0
−1.5
−1.0
−0.5
0.0
0.5
1.0


Log Penalty
35
30
25
20
15
10
5
0
−0.6
−0.4
−0.2
0.0
0.2
0.4